



OPEN 3DFFL: privacy-preserving Federated Few-Shot Learning for 3D point clouds in autonomous vehicles

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This paper presents a comprehensive study of 3D point cloud Federated Few-Shot Learning (3DFFL), focusing on addressing challenges such as limited data availability and privacy concerns in point cloud classification for applications such as autonomous vehicles. We introduce a novel approach that integrates Federated Learning with Few-Shot Learning techniques, with a special emphasis on optimizing network architectures for 3D point cloud data. Our method capitalizes on the strengths of PointNet++ for feature extraction and ProtoNet for classification, all within a federated learning framework to ensure data privacy and collaborative learning. Significantly, the approach is augmented with the use of attention and SoftMax layers, enhancing the feature extraction and classification processes. Extensive experiments on the ModelNet40, ShapeNet, and ScanObjectNN datasets validate our method's accuracy and adaptability in handling 3D point cloud classification, especially in privacy-sensitive and collaborative scenarios. This study not only demonstrates the potential of integrating attention mechanisms and SoftMax layers in 3DFFL but also lays a robust foundation for future advancements in this evolving field, particularly in technologies dependent on 3D data processing.

The field of 3D point cloud classification holds a significant place in several computer vision applications, particularly in emerging areas such as autonomous vehicles and robotics¹. This process involves classifying objects in 3D point clouds into various categories. However, this area faces numerous challenges, especially when considering Federated Few-Shot (FFL) 3D point cloud classification. Additionally, the efficiency of embedding or feature extraction in 3D point clouds typically does not match the levels achieved for 2D image data. This discrepancy largely arises from the dependence on pre-trained models based on images¹, such as ResNet², which are built upon deeply trained networks using extensive image datasets. In contrast, models designed for point clouds, such as PointNet++^{3,4} and PointNet^{5,6}, often rely on less complex networks trained with fewer point cloud samples, characterized by considerable intra-class variation and minimal inter-class distinctions. This variation in data richness and network complexity leads to less effective embedding for 3D point cloud⁷.

The introduction of deep learning techniques has been transformative in the processing of point cloud. Recently, FSL algorithms in the 2D image domain have seen significant strides by creating auxiliary training meta-tasks, employing metric learning or optimization-based methodologies to glean transferable knowledge and disseminate it across new tasks. The potential of these algorithms is particularly promising for enhancing the capabilities of deep networks in the point cloud domain. By utilizing meta-learning techniques⁸ and integrating a limited number of labeled examples, these algorithms aim to boost the generalization ability and mitigate the inherent limitations of traditional deep learning methods in point cloud classification^{9–13}. The motivation behind this research lies in the relatively uncharted territory of FFL on 3D data, especially when compared with its 2D image domain counterpart, necessitating addressing several unique challenging areas such as simultaneous localization and mapping (SLAM) and autonomous vehicles (AVs). These methods depart from traditional approaches that utilize manually crafted features, instead applying deep networks to extract more representative features from either shape projections or raw data points, thereby achieving improved performance. Nevertheless,

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they face two main challenges in the classification of point clouds: the need for a large amount of labeled training data and the difficulty in extending the model's application to new classes or tasks not encountered during the training phase. While techniques such as data augmentation and regularization provide some mitigation against the scarcity of data¹⁴, they often fall short in scenarios lacking substantial labeled training data for specific classes or tasks. The motivation behind this research lies in the relatively uncharted territory of FFSL on 3D data, especially when compared with its 2D image domain counterpart, necessitating addressing several unique challenges:

Challenges in point cloud structures: The inherent nature of point clouds, being irregular, unstructured, and unordered collections of points in Euclidean space¹⁵, presents considerable challenges. The performance and adaptability of current architectures and algorithms in 3D FSL for processing these point clouds remain a topic of ongoing exploration. It is essential to assess their effectiveness concerning the unique properties of point cloud data.

Scarcity of labeled data in 3D learning: There is a significant gap between the effectiveness of 2D FSL algorithms and their 3D counterparts, primarily because 2D algorithms benefit from deeper networks that have been pre-trained on extensive datasets. For example, tieredImageNet¹⁶ encompasses 608 classes with about 800,000 images, in stark contrast to most 3D model datasets, such as ModelNet¹⁷ and ShapeNet¹⁸, which comprise considerably fewer labeled examples. ModelNet includes only 40 classes with a total of 12,311 models. As a result, models trained on these limited datasets tend to yield poorer quality 3D feature clusters, characterized by prominent intra-class variations and minor inter-class distinctions. Developing effective strategies to address these limitations is thus of paramount importance.

To advance the field of 3D Federated FSL (3DFFL), particularly in addressing data scarcity challenges, it is imperative to explore the concept of data sharing among autonomous car manufacturers equipped with LiDAR or similar point cloud generation technology. The utilization of federated learning techniques in this context not only helps mitigate data scarcity issues but also ensures the preservation of data privacy. Furthermore, the integration of FSL into this collaborative approach enhances the capacity to effectively work with limited data resources. This collective effort not only narrows the gap between FSL in the 2D image and 3D data domains but also lays the foundation for the development of precise 3DFFL algorithms tailored specifically for point cloud data, with significant implications for autonomous vehicle applications. This paper represents the first comprehensive study of 3DFFL. Through this comprehensive investigation, we seek to further the understanding and development of 3DFFL techniques, laying a solid foundation for future research and advancements in this evolving domain. The key contributions of this paper are summarized as follows:

- This is the first comprehensive study of Federated Few-Shot Learning for 3D point cloud (3DFFL), with a special focus on optimizing network architectures for 3D point cloud data. This exploration is pivotal for advancing applications like autonomous vehicles, where 3D data processing and model efficiency are critical.
- This study proposed enhancements to the federated few-shot learning method for 3D point clouds, employing attention mechanisms and SoftMax for more effective feature updating.
- Our extensive experiments on ModelNet40, ShapeNet, and ScanObjectNN benchmark datasets, demonstrate the proposed method's effectiveness. These experiments highlight the method's strengths in accuracy and adaptability for 3D point cloud classification, supporting its use in privacy-sensitive and collaborative environments.

Related work

Few-Shot for 3D data

Methods based on deep learning diverge from traditional techniques that rely on manually crafted feature extraction¹⁹, showcasing their ability in deriving complex and nuanced features through the use of extensive network architectures. The techniques employed in handling point cloud data are generally divided into two categories: those that use projection-based methods and those that utilize point-based methods, each differing in their approach to extracting features from either organized grids or raw point data.

Networks using projection-based techniques transform unstructured point cloud data into an organized format, such as through multi-view projections^{20,21}, voxel grids^{22–24}, or lattice structures^{25,26}. Following this, conventional 2D or 3D CNNs are applied to derive features specific to views or structures. Nonetheless, projection-based techniques can face challenges, including potential loss of crucial information or increased demands on memory²⁷.

Contrastingly, point-based methods directly engage with raw point cloud data, bypassing the need to structure them. Techniques such as PointNet, PointNet++, and DGCNN²⁸ facilitate direct learning on these unstructured data sets, eliminating the necessity for voxelization²⁹ or projection. Such point-based methods have been gaining traction due to their efficient handling of irregular and unstructured data in point clouds. Utilizing deep network architectures, these methods, whether projection-based or point-based, are adept at identifying and representing complex patterns within point clouds, offering advancements over traditional feature extraction techniques. Nevertheless, the choice between projection-based and point-based methods should be made with careful consideration of aspects like information loss, memory usage, and the specific characteristics of the point cloud data in question.

In recent developments, there has been an increased focus on applying deep network architectures directly to raw point cloud data. PointNet was a pioneer in this area, utilizing MultiLayer Perceptrons (MLP) for learning individual point features and integrating a symmetric function for global feature aggregation, typically through max-pooling. Following in PointNet's footsteps, subsequent innovations such as PointNet++, PointCNN, RSCNN, DensePoint³⁰, and DGCNN have delved into examining local relationships in specific areas, employing either convolution-based or graph-based networks. These methods have been effective in discerning significant

geometric characteristics, setting new benchmarks in performance. However, the deep learning-based approaches in point cloud processing are somewhat constrained by their reliance on large volumes of annotated data, which limits their adaptability to new, unseen class types. Recent research efforts have been directed towards learning with fewer data samples.

The authors of³¹ have investigated feature representation learning through self-supervision, a distinct approach from few-shot learning (FSL). The study by³² focused on developing a discriminative embedding space for 3D model multi-view images, emphasizing the bias in point cloud shapes. A notable study in this area³³ has been the pioneering systematic analysis of FSL classification of point clouds. In this review, the authors examine the latest networks in 3D point cloud learning and establish robust baselines for addressing this challenge. Furthermore, the authors introduce an innovative, adaptable module, termed the Cross Instance Adaption module, designed to overcome the challenges posed by significant variance within classes and minimal differences between classes in the context of 3D FSL. This module offers the flexibility to be seamlessly incorporated into existing FSL algorithms, thereby enhancing their effectiveness in point cloud scenarios.

Few-Shot for 2D data

As previously discussed, FSL emerges as a critical area in the field of deep learning, notable for its capacity to effectively learn from a limited amount of labeled training data and to generalize well to new tasks. The existing algorithms in FSL are primarily divided into two groups: metric-based and optimization-based methodologies. The metric-based approach is centered on developing an embedding space where pairs of similar samples are positioned in proximity. Notable examples of this include the Matching Network³⁴, which employs a bidirectional LSTM module along with cosine distance for the classification of query samples. Another example is the Prototypical Network^{35,36}, which determines class prototypes by averaging features from the support set and uses squared Euclidean distance to evaluate similarity. Furthermore, Relation Net³⁷ introduces a module capable of learning metrics to calculate relation scores between the support and query sets. In addressing task distribution shifts in Few-Shot Learning (FSL), Luo et al.³⁸ highlight the potential of channel-wise feature transformation to enhance the adaptability and generalization of image representations across varying tasks. Mangla et al.³⁹ introduce the concept of Manifold Mixup, a novel approach aimed at improving the performance of few-shot learning models by encouraging smoother decision boundaries and more generalizable feature representations. Chen et al.⁴⁰ propose the Meta-Baseline method, which simplifies meta-learning for few-shot learning by enhancing a base network's ability to generalize from few examples without complex meta-learning algorithms.

On the other hand, optimization-based methods view meta-learning as a process of optimization. For instance, MAML⁴¹ focuses on learning a set of initialization parameters that are model-agnostic and, through a minimal number of gradient updates, can significantly enhance performance on new tasks. Additionally, MetaOptNet⁴² integrates a differentiable quadratic programming solver into its framework to learn a linear SVM predictor that is feature-relevant, thereby facilitating better generalization to new categories. In our research, we undertake a comprehensive analysis of diverse meta-learning algorithms within the scope of FSL classification for point clouds. We introduce robust baseline methods and specialized components designed to tackle the unique challenges inherent in this area. Our goal is to utilize various meta-learning strategies to improve the efficacy and expand the capabilities of FSL specifically in the context of point cloud data.

Other methods for point cloud processing

In recent times, advancements have been made in 3D shape analysis through the development of methodologies involving view-based and point-based neural networks. Tatarchenko et al.⁴³ introduced Tangent Convolutions, which operate within the tangent space of surfaces to achieve dense prediction in 3D shape analysis. The “Walk in the Cloud” approach⁴⁴ uses curve representations of geometric properties in point clouds to enhance shape analysis. Qian et al.⁴⁵ utilized the PointNet++ structure and the PointNeXt framework to improve training and scaling strategies for 3D point cloud models. These innovations extend the frontier of 3D shape recognition by introducing new geometric processing techniques and optimizing training methodologies. Approaches like View-GCN and MVTN use multi-view 2D projections of 3D shapes for recognition tasks. Wei et al.⁴⁶ proposed View-GCN, achieving state-of-the-art performance in 3D shape classification and retrieval. MVTN⁴⁷ aims to enhance classical multi-view methods by learning the optimal viewpoint for each 3D shape, increasing recognition performance.

Advancements have also been made in point-based approaches. Thomas et al. introduced the KPConv framework, which uses deformable convolutions on point clouds to predict dense 3D data⁴⁸. Wu et al. further developed this concept with PointConv, enabling deep convolutional networks directly on 3D point clouds⁴⁹. FPCConv⁵⁰ addresses challenges in point-based methods through local flattening techniques for point convolution. Recently, there have been efforts to integrate point-based methods with view-based methods. Xiao et al. introduced RegGeoNet, a learning architecture for regular representations of large-scale 3D point clouds, extracting geometric features and learning relationships between them directly⁵¹. Another example is Flattening-Net, which efficiently analyzes equivalent 2D representations from 3D point clouds⁵². However, there are still gaps in these approaches. View-based methods may excel in exploiting 2D image features but often fall short in capturing the complete geometric context of 3D shapes. On the other hand, point-based approaches, while effective in capturing geometric details, can be computationally expensive due to extensive data preprocessing. Thus, integrating multi-view and point-based approaches could offer a holistic solution for the analysis of 3D shapes.

There is also growing interest in combining multi-view approaches with self-supervised learning. PointVST⁵³ method demonstrated self-supervised pre-training can enhance the robustness and generalization of 3D point cloud models. By fine-tuning models from previous work to official implementations, significant improvements were achieved across a wide range of challenging tasks. Additionally, the re-evaluation of the PointNet++

framework using the PointNeXt architecture showcased the potential of advanced training strategies⁴⁵. PointMCD employed multi-view cross-modal distillation to achieve state-of-the-art performance in point cloud encoders⁵⁴. Park et al.⁵⁵ presented a novel, efficient transformer-based architecture for processing 3D point cloud data, achieving faster performance compared to previous transformer based method⁵⁶.

Federated learning for 3D point cloud

Gkillas et al.⁵⁷ propose a federated learning framework for enhancing lidar data resolution in automotive scenes, achieving significant improvements while preserving data privacy by training models collaboratively without sharing raw data. Similarly, Zhang et al.⁵⁸ introduce a reconfigurable holographic surface aided collaborative wireless SLAM system using federated learning for autonomous driving, enhancing localization and mapping accuracy while maintaining data privacy through decentralized model training. Moreover, Lan et al.⁵⁹ present a decentralized federated learning framework utilizing BEV feature exchange pyramid networks for 3D object detection in small and distant situations, improving detection accuracy while preserving data privacy by avoiding centralized data aggregation. These studies explore various point cloud processing techniques; however, to the best of our knowledge, no one has addressed Federated Few-Shot Learning for 3D point cloud classification, making it a relevant area for further investigation.

Methodology

In this section, we detail the methodology employed in our study, outlining the integrated approach of FL and FSL applied to 3D point cloud classification.

Loss function with FL adaptation

Incorporating FL into the proposed method necessitates an adaptation of the loss function to suit the distributed nature of FL. The loss calculation in our FL-based approach consists of two primary components: the embedding loss and the learnable loss, both of which are computed locally at each node and then aggregated to update the global model.

Federated embedding loss: The embedding loss at each federated node is calculated using cross-entropy loss. It measures the discrepancy between the predicted class scores ($\mathbf{Y}$) and the ground truth labels ($\mathbf{L}$) for a local set of samples at the node. The embedding loss at a node is given by:

$$L_{embedding}^{(node)} = -\frac{1}{N^{(node)}} \sum_{i=1}^{N^{(node)}} \sum_{c=1}^C \mathbf{L}_i^c \log(\mathbf{Y}_i^c), \quad (1)$$

where: - $N^{(node)}$ represents the number of samples at the node. - C is the total number of classes. - $\mathbf{L}_i^c$ and $\mathbf{Y}_i^c$ are the ground truth label and predicted class score for the i -th sample and c -th class at the node.

Federated learnable loss: The learnable loss at each node leverages the concept of local prototypes to measure the dissimilarity between local query samples ($\mathbf{Q}^{(node)}$) and local prototype vectors ($\mathbf{P}^{(node)}$). It is computed as:

$$L_{learnable}^{(node)} = -\frac{1}{N^{(node)}} \sum_{i=1}^{N^{(node)}} \log \left(\frac{\exp(-d(\mathbf{P}_{L_i}^{(node)}, \mathbf{Q}_i^{(node)}))}{\sum_{c=1}^C \exp(-d(\mathbf{P}_c^{(node)}, \mathbf{Q}_i^{(node)}))} \right). \quad (2)$$

Overall loss: The overall loss at each federated node combines the local embedding loss and local learnable loss. This local overall loss is then aggregated across all nodes to update the global model. The overall loss at a node is:

$$L_{overall}^{(node)} = \lambda_1 \cdot L_{embedding}^{(node)} + \lambda_2 \cdot L_{learnable}^{(node)} + \lambda_3 \cdot L_{comp}^{(node)}, \quad (3)$$

where λ_1 and λ_2 are hyperparameters, and $L_{comp}^{(node)}$ represents the loss for computational components utilized by the node's local network. By minimizing the overall loss in a federated manner, the model optimizes both embedding and learnable components across the distributed network, leading to enhanced performance in FSL for 3D point cloud classification. This federated adaptation of the loss function ensures that learning is effectively distributed across nodes, each contributing to the refinement of the global model while preserving data privacy and locality.

Overall objective

When integrating FL into the overall objective of the feature extraction and learning method, the goal is to minimize the combined loss across the distributed network while optimizing the model's parameters. This involves adapting the objective to suit the federated framework, where each node contributes to the learning process without directly sharing data. The overall objective in a federated setting becomes a collaborative effort to optimize a global model through local computations and aggregated updates. The mathematical expression for the federated overall objective is:

$$\min_{\theta} \left(\sum_{k=1}^K w_k \cdot \left(\lambda_1 \cdot L_{embedding}^{(k)} + \lambda_2 \cdot L_{learnable}^{(k)} + \lambda_3 \cdot L_{comp}^{(k)} \right) \right), \quad (4)$$

In this context, θ represents the global parameters of the combined model, which include both the embedding and learnable components. The embedding loss calculated at the k -th federated node is denoted by $L_{embedding}^{(k)}$, while $L_{learnable}^{(k)}$ denotes the learnable loss at the k -th node. Additionally, $L_{comp}^{(k)}$ represents the loss for additional computational components at the k -th node. The hyperparameters λ_1 , λ_2 , and λ_3 control the relative weights of

the respective loss terms. The weight assigned to the k -th node's contribution, w_k , could be based on the number of samples or the computational power of the node. Lastly, K is the total number of nodes participating in the FL process. The overall objective emphasizes the collaborative minimization of loss across all nodes. Each node computes its local loss and updates its parameters, which are then aggregated to update the global model parameters θ . This process ensures that the global model benefits from diverse local insights while maintaining data privacy and efficiency in learning. The federated approach is particularly effective in scenarios with distributed data sources, enabling the model to learn generalized features and prototypes that are robust for FSL based 3D point cloud classification tasks.

FSL

In this section, we describe the methodology for our 3DFFSL, integrating FL with PointNet++ (embedding) and ProtoNet (learnable) components. The challenge lies in developing a model capable of accurately classifying new point clouds into their respective classes, despite having limited labeled examples for each class. This model, leveraging the FL framework, enables learning from various datasets while maintaining privacy and data diversity without the need for data sharing. The model aims to capture discriminative features from the input point clouds effectively. It incorporates an attention mechanism to highlight relevant information, crucial for dealing with the sparsity and irregularity of point clouds. The FL approach adds a layer of complexity and innovation to this process. By distributing the learning process across multiple nodes, the model benefits from a wider range of data variances and real-world scenarios, which is particularly advantageous in FSL settings where data scarcity is a common issue.

Additionally, the learnable component for classification, represented by ProtoNet, is adapted to work within the federated framework. This adaptation ensures that the model not only learns generalized features across distributed datasets but also retains the ability to perform robustly on unseen point clouds. The ultimate objective is to achieve accurate and robust classification performance on new point clouds, effectively overcoming the inherent challenges of FSL in the 3D domain, while also benefiting from the collaborative and privacy-preserving aspects of FL.

FL adaptation in embeddings

The PointNet++ architecture, an advanced extension of the original PointNet designed for point cloud analysis, is further enhanced by incorporating FL techniques. This integration is crucial in addressing the challenges of data scarcity while ensuring data privacy and leveraging the diversity of data sources. PointNet++ operates across multiple nodes (e.g., devices or servers), each with its own subset of data. The architecture's hierarchical set abstraction (SA) and feature propagation (FP) layers are adapted to function in a distributed manner.

Federated SA Layer: Each node receives a subset of the input point set $\mathbf{X}$ and applies the SA layer locally. This process involves partitioning local point sets into smaller regions $S = \{S_1, S_2, \dots, S_M\}$, where M represents the number of regions formed within each node. A shared multi-layer perceptron (MLP) learns local features for these regions. The output at each node is a transformed set of points $\mathbf{X}'$ and their corresponding features $\mathbf{F}$.

Federated FP Layer: Similar to the SA layer, the FP layer is executed locally at each node. It refines the features of the transformed points by aggregating information across different scales. The local outputs are refined feature representations $\mathbf{F}'$.

Following the local processing, the key innovation in the federated adaptation lies in the aggregation of learned features across nodes. After each training epoch, the model parameters or the learned feature representations from each node are sent to a central server. This server performs an aggregation operation, such as Federated Averaging, to update the global model. The updated global model is then redistributed to each node for further training.

This federated approach to embeddings allows PointNet++ to benefit from diverse datasets without compromising data privacy. Each node contributes to the learning process with its local data, ensuring that the global model gains a comprehensive understanding of various point cloud patterns. This method is particularly advantageous in scenarios with data that cannot be centralized due to privacy or logistical constraints.

The mathematical foundation of PointNet++ remains intact in this federated setting. However, the FL adaptation ensures that the learning process is collaborative, and privacy-preserving. The global model, enriched with diverse data insights, becomes more robust and generalizable, making it well-suited for real-world applications where data is inherently distributed and varied. Integrating FL with ProtoNet for FSL introduces a novel approach to learning embeddings in a privacy-preserving manner. In this adapted version, the process of learning and utilizing prototypes for classification is distributed across multiple nodes in the federated network.

Federated prototype computation: In the FL setting, each node in the network holds a subset of the support examples for each class c , denoted as $S_c^{(node)}$. The prototype computation is performed locally at each node:

$$\mathbf{p}_c^{(node)} = \frac{1}{|S_c^{(node)}|} \sum_{\mathbf{x}_{ci} \in S_c^{(node)}} \mathbf{x}_{ci}, \quad (5)$$

where $\mathbf{p}_c^{(node)}$ is the local prototype for class c at a specific node, and $|S_c^{(node)}|$ is the number of support examples for class c at that node. The local prototypes are then aggregated to form the global prototypes. This aggregation can be done using techniques such as Federated Averaging, ensuring that the global prototypes $\mathbf{p}_c$ represent the mean embedding of support examples from class c across all nodes.

Federated classification: In the federated context, query examples may be distributed across different nodes. Each node computes the distances between its local query examples and the global prototypes. The classification of a query example $\mathbf{q}$ at a node is based on the distance to these global prototypes:

$$\hat{y}^{(node)} = \arg \min_c d(\mathbf{q}, \mathbf{p}_c), \quad (6)$$

where $\hat{y}^{(node)}$ represents the predicted class label for the query example $\mathbf{q}$ at that node.

The integration of FL into the embedding learning process with ProtoNet enables the model to leverage the diverse data sources effectively. Each node contributes to the formation of more robust and representative global prototypes by computing local prototypes based on its available data. This approach not only preserves data privacy but also enhances the model's ability to generalize across varied datasets. Consequently, the federated version of ProtoNet becomes more adaptable and efficient in scenarios where data cannot be centralized or where privacy concerns are paramount.

Enhanced FL in transfer of features

Incorporating FL into the feature transfer process involves a nuanced adaptation of the attention mechanism to suit a collaborative learning environment. This adaptation aims to optimize the feature extraction and classification process across a network of distributed nodes, each possessing unique subsets of data, thus enhancing the model's overall performance and robustness.

Attention mechanism: In a federated setup, each participating node computes its local query features ($\mathbf{Q}^{(node)}$) and prototype features ($\mathbf{P}^{(node)}$). The attention mechanism is tailored to operate at the node level:

Attention weight calculation: Each node independently calculates attention weights for its local features. This calculation considers the specific characteristics and distribution of the data at the node, allowing the mechanism to adaptively emphasize the most informative parts of the local dataset.

$$\mathbf{A}_Q^{(node)}, \mathbf{A}_P^{(node)} = \text{AttentionWeights}(\mathbf{Q}^{(node)}, \mathbf{P}^{(node)}). \quad (7)$$

SoftMax application: The local attention weights ($\mathbf{A}_Q^{(node)}, \mathbf{A}_P^{(node)}$) are then normalized using the SoftMax function. The attention-weighted features are derived by element-wise multiplication of these normalized weights with the local features.

$$\mathbf{Q}'^{(node)} = \mathbf{Q}^{(node)} \odot \text{SoftMax}(\mathbf{A}_Q^{(node)}), \quad (8)$$

$$\mathbf{P}'^{(node)} = \mathbf{P}^{(node)} \odot \text{SoftMax}(\mathbf{A}_P^{(node)}). \quad (9)$$

Updating local features: The local attention-weighted features are then combined with the original features to obtain the enhanced feature set. This step is crucial as it integrates the selective emphasis of the attention mechanism with the raw feature set, leading to a more informative representation.

$$\mathbf{Q}''^{(node)} = \mathbf{Q}'^{(node)} + \mathbf{A}_Q^{(node)}, \quad (10)$$

$$\mathbf{P}''^{(node)} = \mathbf{P}'^{(node)} + \mathbf{A}_P^{(node)}. \quad (11)$$

Aggregation of federated features: Upon local feature enhancement, the model undergoes a phase of global feature aggregation. This process is pivotal for FL, as it consolidates the diverse insights gained from individual nodes:

Feature sharing and aggregation: Each node in the federated network computes its local feature representations and then periodically shares these with a central aggregator. Let's denote the feature representation at node i as $\mathbf{F}^{(i)}$. The aggregator then performs federated averaging across all nodes to update the global model. Mathematically, this can be represented as follows:

Let N be the total number of nodes in the federated network. The aggregated global feature representation $\mathbf{F}_{global}$ is computed as the average of the local feature representations from all nodes:

$$\mathbf{F}_{global} = \frac{1}{N} \sum_{i=1}^N \mathbf{F}^{(i)}. \quad (12)$$

This equation represents how the global model aggregates local insights by averaging the feature representations from each node.

Redistribution of global model

Once the global feature representation $\mathbf{F}_{global}$ is computed, it is redistributed back to all nodes in the network. This redistribution process ensures that each node in the federated network can update its local model with the newly aggregated global insights. The updated local model at node i after receiving the global feature representation can be represented as:

$$\mathbf{F}_{updated}^{(i)} = \mathbf{F}_{global}, \quad (13)$$

where $\mathbf{F}_{updated}^{(i)}$ represents the updated local feature representation at node i after incorporating the global model's insights. This redistribution is crucial as it allows each node to benefit from the learnings and discoveries made by other nodes, thereby enhancing the overall learning and predictive performance of the federated network.

By continuously iterating through these steps of local enhancement, global aggregation, and redistribution, the FL model progressively improves, becoming more robust and accurate over time.

This enhanced federated approach in the feature transfer stage ensures a synergistic combination of local learning with global model enhancement. The localized attention mechanism allows for nuanced feature extraction tailored to each node's data, while the global aggregation phase ensures that these individual insights contribute to the collective intelligence of the model. Consequently, the model not only exhibits improved performance in 3DFFL tasks but also displays heightened adaptability and generalizability, essential for real-world applications where data diversity and privacy are paramount.

Figure 1 presents the 3DFFL framework employed for collaborative training in the context of 3D point cloud processing for autonomous vehicles. Each Vehicle Package (VP) represents a self-contained unit that includes an autonomous vehicle equipped with 3D data acquisition tools like LiDAR and a communication system for data relay. The raw point cloud data collected by the VPs undergo preprocessing and validation at the respective car company's headquarters (HQ) i.e., node. These HQs are responsible for the initial training of local model weights, ensuring the data's integrity and suitability for the global learning process. The locally trained weights are then securely transmitted to a central server, where they are aggregated to enhance a shared global model. This aggregation process allows for collaborative learning across different entities while maintaining the privacy of each participant's data. The central server then distributes the updated global model weights back to each HQ, facilitating continuous improvements in local models and preserving the confidentiality of the data. This iterative process ensures that the global model benefits from diverse, privacy-conscious contributions, advancing the performance and reliability of autonomous driving technologies through 3DFFL. Moreover, an overview of the proposed model used for few shot is presented in Fig. 2. The extracted features of query and prototype features are enhanced through the incorporation of an attention mechanism. This attention mechanism selectively emphasizes relevant information, followed by the application of a SoftMax function to the attention features. The attention features are multiplied by the SoftMax output and added to the attention features, resulting in updated features. These updated features are then transferred to the learnable ProtoNet for point cloud class prediction. The attention mechanism offers benefits such as selective emphasis, enhancing the model's ability to capture discriminative patterns, and improving the accuracy of point cloud classification.

To further clarify the training process of our model, we present Algorithm 1 for the Federated Few-Shot Learning for 3D Point Clouds in Autonomous Vehicles. This algorithm describes the steps for local model adaptation on edge devices, privacy-preserving techniques, aggregation of model updates on the server, and redistribution of the global model.

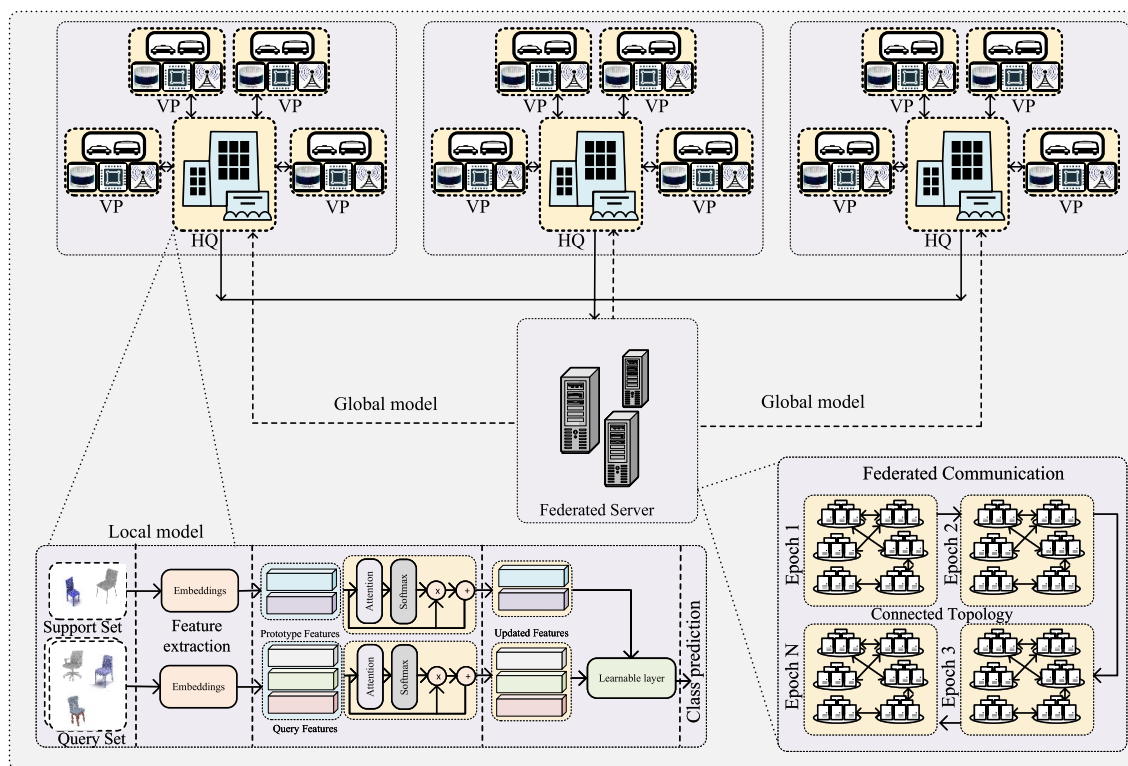


Figure 1. Collaborative training and data privacy in 3DFFL, depicting the process of autonomous vehicles (VPs) collecting data, local HQs performing preprocessing and initial model training, and a central server aggregating and redistributing updated global model weights to enhance classification while preserving data integrity.

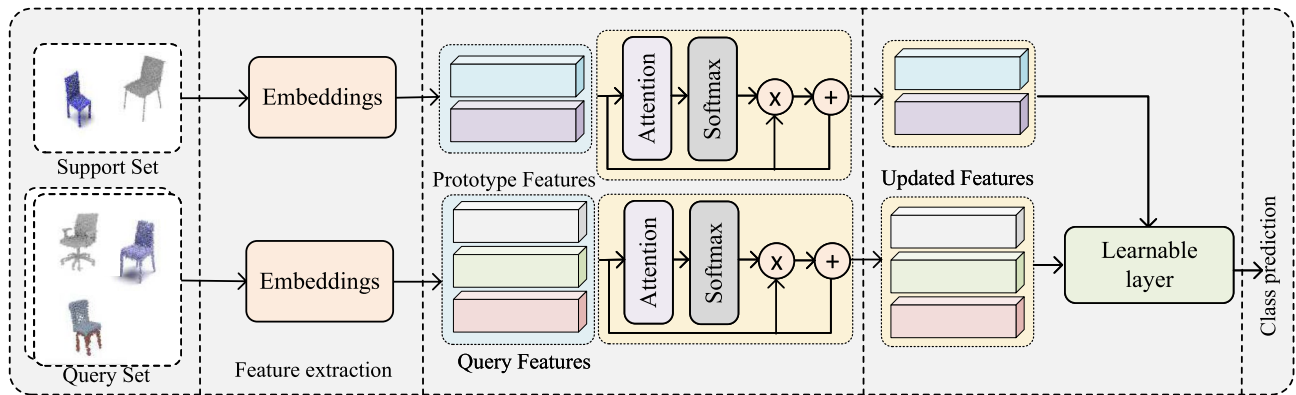


Figure 2. Overview of the proposed approach. The proposed approach enhances the extracted query and prototype features using an attention mechanism. Relevant information is selectively emphasized, and SoftMax is applied to the attention features. The resulting multiplication and addition yield updated features, which are used for point cloud class prediction. The attention mechanism improves discriminative pattern capture, enhancing accuracy in point cloud classification.

Require: Number of communication rounds M , local datasets $D_i = X_i, Y_i$ for $i = 1 \dots N$
Ensure: Global model for few-shot learning

```

1: for  $t = 1$  to  $M$  do
2:   // Edge device side: Local model adaptation
3:   for  $i = 1$  to  $N$  do
4:     // Few-shot learning adaptation
5:     Adapt local model to perform few-shot learning using available data
6:     Update local model parameters using the adapted learning process
7:
8:     // Privacy-preserving adaptation
9:     Perform privacy-preserving techniques (e.g., differential privacy) on local data if required
10:  end for
11:
12:  // Server side: Aggregation and model update
13:  Aggregate local model updates from all devices
14:  Update global model using the aggregated updates
15:
16:  // Privacy-preserving model redistribution
17:  Distribute the updated global model to all devices
18: end for

```

Algorithm 1. Federated training process.

Handling data heterogeneity in Federated Few-Shot Learning

In a federated learning setup, the presence of data heterogeneity among different nodes presents significant challenges to model performance and generalization. The heterogeneity at the node level can lead to instances of localized overfitting, thereby decreasing the effectiveness of global modeling due to variations in data distribution. This heterogeneity arises from differences in data collection methods, environmental conditions, and specific use cases across the nodes. Consequently, the model may perform well on data similar to that of one node but struggle with data from another node. Although exposure to diverse data can enhance model generalization, significant differences can hinder the learning process and impair the model's capability for global tasks. The Federated Averaging (FedAvg) approach addresses these discrepancies in data distributions by aggregating model updates from all participants. However, this method may not suffice for highly heterogeneous data. Thus, we employ data normalization techniques to standardize the data across nodes, thereby reducing variability. This approach contributes to simulating various scenarios that facilitate robust feature learning by the model. Additionally, adaptive learning rates offer a means of adjusting the learning rate based on the distribution of local data, thereby preventing overfitting to localized data and promoting better generalization.

Experiments, results and discussion

Datasets

We adopted specialized splits for the ModelNet40 and ShapeNet datasets³³, tailored for FSL-based 3D point cloud classification. These splits were chosen with an emphasis on maintaining similar data distributions between the training and testing sets. Specifically, the ModelNet40 dataset is split into 30 training classes with 9240 instances and 10 testing classes comprising 3104 instances. Conversely, the ShapeNet dataset is more extensive, encompassing a total of 30,073 instances. It is divided into 50 training classes with 21,722 samples and 20 testing classes holding 8351 samples. For the ScanObjectNN dataset, the splits are constructed to further challenge the model due to real-world data complexity. The ScanObjectNN⁶⁰ consists of 15 categories with a total of 14,298 perturbed objects. This dataset is divided into three non-overlapping subsets, each containing five classes. The splits are as

follows: Split 0 has 9509 instances for training and 4789 for testing, Split 1 has 9129 training instances and 5169 testing instances, and Split 2 includes 9958 training instances and 4340 testing instances. For federated training, the training set is divided into three random parts, as illustrated in Fig. 1. Detailed descriptions of these dataset splits can be found in^{33,60}. Figure 3 presents selected examples from the ModelNet40, and ShapeNet datasets, offering a glimpse into the variety and nature of the point clouds utilized in our research.

Testing and training

As mentioned earlier, to ensure a standardized and equitable evaluation framework, we implement the widely-recognized episode-based FSL methodology^{33,60} throughout our training and assessment phases. The training regimen entails developing the network from its foundational stages over 80 epochs. Within each epoch, the network undergoes 400 meta-training episodes, supplemented by 600 validation episodes randomly selected from the training dataset to track the network's development. The episode-based FSL paradigm structures each episode to include N distinct classes, with each class represented by K labeled support instances and Q query instances. This configuration is often termed as the N - way K - shot Q - query setting, facilitating the network's learning process from a constrained set of labeled examples and enabling it to generalize to query instances. Upon completing the meta-training phase, the network undergoes an evaluation phase using 700 meta-testing episodes, randomly chosen from the test dataset. These episodes adhere to the identical N -way K -shot Q -query format. The performance is assessed by averaging the classification outcomes across these meta-testing episodes. Additionally, we incorporate a 5-fold cross-validation technique. This approach involves partitioning the training dataset into five equal segments, with each segment serving once as a validation group while the others are utilized for training. This cycle is repeated five times, and the aggregate of these results offers a comprehensive and reliable measure of the network's effectiveness on the test dataset. By following these standardized procedures and incorporating cross-validation, we ensure reliable and unbiased evaluation of our 3DFFL, reporting the average performance across multiple estimations.

For experimental purposes, we chose to utilize three client nodes, dividing the datasets into three equal parts. It should be noted that we used the splits provided by Ye et al.^{33,60}. These splits are disjoint in nature, making them significantly more challenging compared to natural splits of the dataset. This partitioning ensures diversity in the data seen by each client, promoting robust model training. Each node employs a continuous training method with a patience of five epochs to ensure stability and convergence. The federated learning process involves multiple communication rounds between the clients and the central server. Specifically, we conducted 80 training rounds, with each client performing local training for 40 epochs per round. We used the FedAvg algorithm to aggregate the local models into a global model. This iterative process continued until the global model converged. By following this approach, we set a baseline for few-shot based federated 3D classification of point clouds.

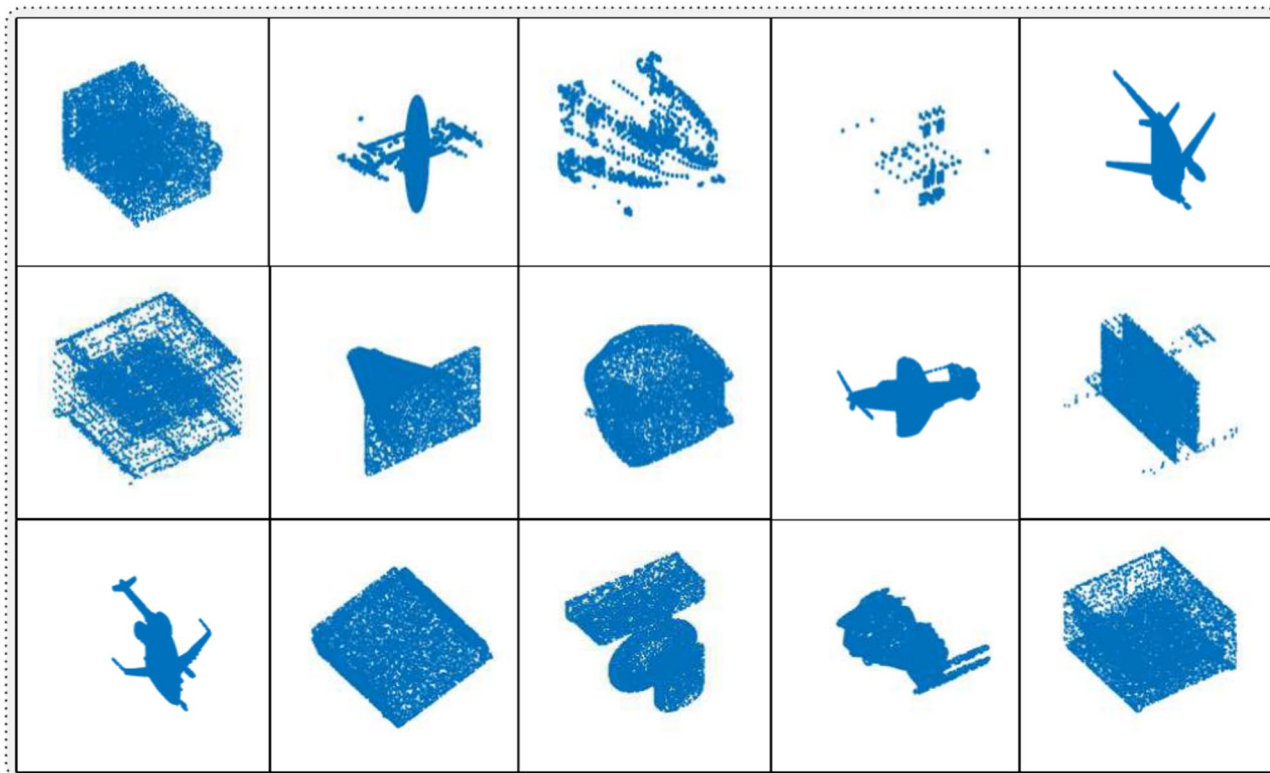


Figure 3. The point clouds used in this study are a subset of samples selected from the ModelNet40 and ShapeNet datasets. These point clouds have undergone normalization and resampling processes to ensure consistency in their scale and point density.

Upon completing the meta-training phase, the network undergoes an evaluation phase using 700 meta-testing episodes, randomly chosen from the test dataset. These episodes adhere to the identical $N - way K - shot Q - query$ format. The performance is assessed by averaging the classification outcomes across these meta-testing episodes. Additionally, we incorporate a 5-fold cross-validation technique. This approach involves partitioning the training dataset into five equal segments, with each segment serving once as a validation group while the others are utilized for training. This cycle is repeated five times, and the aggregate of these results offers a comprehensive and reliable measure of the network's effectiveness on the test dataset. By following these standardized procedures and incorporating cross-validation, we ensure reliable and unbiased evaluation of our 3DFFL, reporting the average performance across multiple estimations.

Implementation details

In our implementation, we utilize PointNet++ as the feature extraction network i.e., the backbone. PointNet++ is a deep learning architecture specifically designed for processing and understanding point cloud data. Unlike traditional methods that rely on handcrafted features or grid-based representations, PointNet++ directly operates on the raw point cloud input. The key innovation of PointNet++ lies in its ability to learn point-wise features and capture global context simultaneously. It achieves this by employing shared MLP to extract local features from individual points and then using a max-pooling operation to aggregate the point features into a global feature vector. This global feature vector captures the overall characteristics of the entire point cloud. By leveraging a symmetric function, PointNet++ ensures that the learned representation remains invariant to point order permutations. This property allows the network to effectively handle unordered and unstructured point cloud data. PointNet++ has shown remarkable performance in various 3D tasks, including classification, segmentation, and object recognition. It is a pioneering work in the field of deep learning for point cloud processing and has paved the way for subsequent advancements and architectures that build upon its foundations. We employ the Adam optimizer with an initial learning rate of 0.00008 and a gamma value of 0.5. The learning rate is reduced every 5 epochs to facilitate effective training. To augment the training data, we apply random points jittering and rotation techniques, similar to the approach used in⁶. These augmentation strategies help introduce variability and enhance the model's ability to generalize.

Comparison with state of the art

The results and comparison provided in Table 1 present the performance of different methods, including state-of-the-art approaches, on the ModelNet40 and ShapeNet datasets. Table 1 shows the classification accuracy for each method under different settings, such as 5-way 1-shot and 5-way 5-shot.

Upon analyzing the results, it is apparent that our proposed 3DFFL achieves competitive performance compared to other state-of-the-art methods, while also ensuring privacy during the collaborative training process. In the ModelNet40 dataset, our method demonstrates comparable accuracy, with a score of 73.8% for 5-way 1-shot and 87.55% for 5-way 5-shot. While these results are slightly lower than the highest scores achieved by other methods, it is important to note that our approach offers significant advantages in terms of privacy preservation due to its FL framework. In the context of the ShapeNet dataset, our method's performance, with an accuracy of 72.81% for 5-way 1-shot and 82.68% for 5-way 5-shot, is noteworthy. Although not surpassing all other methods in every metric, the results are still commendable considering the privacy-preserving nature of our FL approach. This aspect is particularly vital in scenarios involving multiple entities, such as different automotive companies, where data privacy is paramount.

Method	ModelNet40		ShapeNet	
	5-Way 1-Shot	5-Way 5-Shot	5-Way 1-Shot	5-Way 5-Shot
ProtoNet ³⁶	77.55	82.95	71.75	78.25
GGCN ⁶¹	38.56	43.35	39.56	41.39
Self Supervised FSL ³¹	53.15	61.63	52.26	57.13
3D CapsNet ⁶²	32.39	39.19	29.68	34.29
Point-BERT ⁶³	69.41	86.83	73.92	82.86
PointGPT ⁶⁴	76.12	88.92	72.12	82.41
MetaOptaNet ⁴²	67.05	85.05	68.27	81.06
Feng et al. ⁶⁵	61.36	73.2	65.09	75.89
Cascade Graph Neural Networks ⁶⁶	59.91	71.45	63.91	75.42
Luo et al. ³⁸	71.44	86.78	69.19	83.37
Mangla et al. ³⁹	69.73	83.25	68.53	79.71
Chen et al. ⁴⁰	71.33	85.27	70.16	81.08
Ye et al. ³³	75.70	87.15	73.57	83.24
Proposed (3DFFL)	73.80	87.55	72.81	82.68

Table 1. Comparison of our proposed method with state-of-the-art methods on the ModelNet40 and ShapeNet datasets. The table presents the classification accuracy under different settings, including 5-way 1-shot and 5-way 5-shot. The bold text indicate the highest values. Significant values are in bold.

These promising results indicate the effectiveness and competitiveness of our proposed method in 3DFFL based classification. The utilization of attention mechanisms and the incorporation of PointNet++ and ProtoNet contribute to the improved accuracy achieved. The attention mechanism allows our model to selectively emphasize relevant information, enhancing discriminative feature extraction and classification. By leveraging the strengths of PointNet++ and ProtoNet, our method effectively captures the distinctive characteristics of point clouds and enables accurate classification with limited labeled examples. The results of our method demonstrate its potential in advancing the field of few-shot point cloud classification. The superior performance achieved in comparison to state-of-the-art methods showcases the effectiveness of our approach and its ability to handle the challenges of few-shot learning in the context of 3D data.

The results presented in Table 2 demonstrate the performance of our proposed method compared to state-of-the-art approaches on the ScanObjectNN dataset. The evaluation was conducted across three distinct splits for both 5-way 1-shot and 5-way 5-shot scenarios. The proposed method achieved competitive results, consistently performing well across different splits, and demonstrating robustness and reliability. This indicates the potential of our method for practical applications, especially considering its strong results despite the inherent challenges of the ScanObjectNN dataset. Additionally, our model can deliver results on real-world datasets while ensuring data privacy. In our evaluations, the performance of 3DFFL is comparable to that of Ye et al.³³ in various settings, as detailed in Table 1. Despite minor performance variations, the primary distinction of our approach lies in its robust privacy-preserving capabilities, which are not addressed by other studies. This is significant for applications involving sensitive data, where maintaining confidentiality is as crucial as achieving high accuracy. Our method successfully demonstrates that it is possible to achieve competitive accuracy while also ensuring stringent data privacy across distributed nodes.

Hyperparameter tuning and analysis

To determine the optimal values for the hyperparameters λ_1 , λ_2 , and λ_3 , we employed a grid search method. This approach, although computationally intensive, ensures that we can identify the most effective combination of hyperparameters for our model. The grid search was conducted over a range of values from 0.1 to 1.0 for each hyperparameter. This exhaustive search, despite its high resource demands, provided a thorough exploration of the hyperparameter space. Figure 4a,b illustrate the results of the hyperparameter tuning for the ModelNet40 and ShapeNet datasets, respectively. Each point in the 3D scatter plots represents a specific combination of λ_1 , λ_2 , and λ_3 , with the color indicating the accuracy achieved. Upon careful analysis of the results from both datasets, we determined different sets of optimal hyperparameters for each. For the ModelNet40 dataset, the selected values are $\lambda_1 = 0.6$, $\lambda_2 = 0.8$, and $\lambda_3 = 0.9$. For the ShapeNet dataset, the optimal values are $\lambda_1 = 0.5$, $\lambda_2 = 0.7$, and $\lambda_3 = 0.8$. These values provided robust performance on their respective datasets, achieving high accuracy while maintaining consistency.

This method, while not ideal due to its computational and time demands, provided us with crucial insights into the sensitivity of our model's performance to these hyperparameters. The results underscore the importance of careful hyperparameter tuning in achieving optimal performance in federated learning scenarios.

Comparison with baseline methods

To provide a comprehensive evaluation of our proposed 3DFFL method, we conducted a detailed comparison with several baseline methods. By “baseline,” we refer to the standard/legacy methods used for few-shot learning (FSL). These methods represent the state-of-the-art in FSL and point cloud classification. ProtoNet³⁵ employs a prototypical network that computes distances from class prototypes to effectively classify new instances, implemented directly to assess its out-of-the-box performance on 3D point cloud data. RelationNet³⁷ uses a relational module to learn deep distance metrics, enabling it to handle various types of data without pre-training. GNN⁶⁷ leverages the capabilities of graph neural networks, particularly suitable for data that naturally forms graphs such as 3D point clouds. Meta Learner⁶⁸ adapts quickly to new tasks using a meta-learning approach, and MetaNet⁴² focuses on rapid adaptation through fast parameterization. These methods, along with combinations of these

Method	ScanObjectNN					
	5-way 1-shot			5-way 5-shot		
	Split 1	Split 2	Split 3	Split 1	Split 2	Split 3
ProtoNet ³⁶	45.10	44.10	49.64	63.91	64.43	63.49
Point-BERT ⁶³	33.22	33.92	38.17	50.32	49.15	53.55
MetaOptaNet ⁴²	45.98	48.48	58.33	64.13	68.27	71.82
Feng et al. ⁶⁵	26.43	46.37	49.02	56.10	60.83	52.68
Cascade GNN ⁶⁶	38.32	41.24	51.45	52.74	58.81	61.87
Luo et al. ³⁸	45.63	44.22	54.89	63.87	66.07	74.74
Mangla et al. ³⁹	43.15	44.39	64.60	61.29	65.37	59.52
Chen et al. ⁴⁰	45.75	47.41	56.77	59.35	67.52	67.62
Ye et al. ³³	48.61	57.96	62.16	65.96	73.82	73.11
Proposed (3DFFL)	45.23	51.44	59.12	59.28	69.12	69.23

Table 2. Evaluation results on ScanObjectNN dataset. Significant values are in bold.

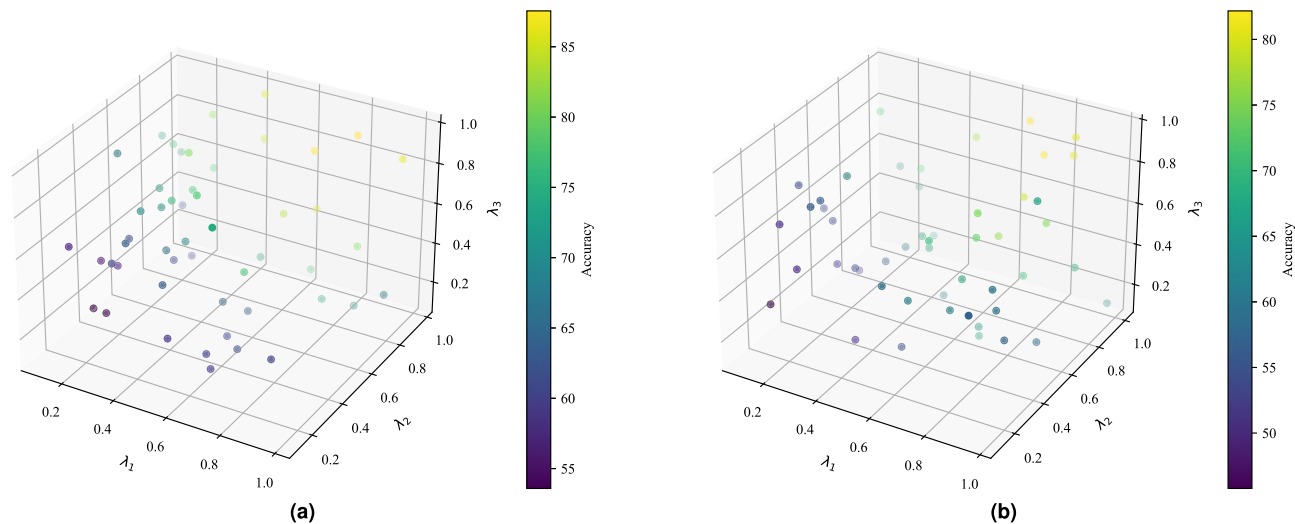


Figure 4. Hyperparameter tuning results for (a) ModelNet40 and (b) ShapeNet datasets. Zoom in for a better view.

frameworks with various point cloud processing architectures such as PointNet, PointCNN, and PointNet++, were directly implemented without any backbones to evaluate their basic performance. The metric used for evaluating these methods is accuracy, particularly in the settings of 5-Way 1-Shot and 5-Way 5-Shot. The results, as shown in Table 3, present a comparative analysis of our proposed 3DFFL method against these established baseline methods. Our 3DFFL method demonstrates superior performance across both the ModelNet40 and ShapeNet datasets. Specifically, our method achieves an accuracy of 73.8% and 87.55% in the 5-Way 1-Shot and 5-Way 5-Shot settings, respectively, on the ModelNet40 dataset, and 72.81% and 82.68% on the ShapeNet dataset, outperforming other approaches in these accuracy metrics.

These results emphasize the efficacy of the 3DFFL method, particularly in the context of Federated Few-Shot Learning. The superiority of our method in these benchmark datasets demonstrates its robustness and adaptability to different 3D point cloud classification tasks. Moreover, the integration of FL principles into the FSL paradigm not only enhances performance but also ensures data privacy and collaborative learning among various entities, which is crucial in fields such as autonomous vehicle development. This blend of high accuracy and privacy-preserving features positions our 3DFFL method as a leading solution in the domain of point cloud classification for real-world applications.

Ablation study

The results presented in Table 4 stem from an ablation study conducted to evaluate the performance improvements brought by the addition of attention and SoftMax layers in various FSL models. In this study, we specifically focused on the impact of these layers on the accuracy of different backbone architectures, including PointNet⁶, PointNet++⁴, and DGCNN⁷⁰, when combined with ProtoNet and Relationnet methods.

The check marks in the table indicate the presence of either the attention or SoftMax layer or both. It is evident from the data that the inclusion of these layers, particularly in combination, has a significant impact on the model's performance. For instance, in the PointNet++ architecture combined with ProtoNet, the incorporation of both attention and SoftMax layers leads to the highest accuracy scores of 73.80% and 87.55% in the 5-Way

Method	Datasets			
	ModelNet40		ShapeNet	
	5-Way 1-Shot	5-Way 5-Shot	5-Way 1-Shot	5-Way 5-Shot
ProtoNet ³⁵	65.31	79.04	65.96	78.77
RelationNet ³⁷	64.11	75.75	65.88	76.25
GNN ⁶⁷	59.69	76.06	64.98	76.14
Meta Learner ⁶⁸	58.69	76.60	62.64	73.10
MetaNet ⁴²	64.99	79.54	65.08	77.81
PointNet ⁶ + ProtoNet	65.31	79.04	65.96	78.77
PointCNN ⁶⁹ + ProtoNet	60.38	76.95	64.02	76.34
PointNet++ ⁴ + ProtoNet	64.96	83.66	66.33	80.95
Proposed (3DFFL)	73.8	87.55	72.81	82.68

Table 3. Comparative analysis of FSL methods and baselines on ModelNet40 and ShapeNet datasets. Significant values are in bold.

Backbone	Method	A	S	ModelNet40	
				5-Way 1-Shot	5-Way 5-Shot
PointNet ⁶	ProtoNet	✓		55.69	74.02
			✓	62.37	79.58
		✓	✓	57.86	86.16
	Relationnet		✓	55.54	75.29
		✓		50.67	54.84
		✓	✓	59.59	74.16
PointNet++ ⁴	ProtoNet	✓		51.23	34.06
			✓	69.46	65.65
		✓	✓	73.80	87.55
	Relationnet	✓		49.06	84.10
			✓	65.44	58.50
		✓	✓	54.89	52.01
DGCNN ⁷⁰	ProtoNet	✓		56.47	42.92
			✓	70.81	73.78
		✓	✓	36.52	54.52
	Relationnet		✓	37.17	61.32
		✓		34.48	28.04
		✓	✓	67.08	64.30

Table 4. Ablation study on the impact of Attention and SoftMax layers in FSL models on ModelNet40 datasets. Significant values are in bold.

1-Shot and 5-Way 5-Shot settings, respectively. This result is a substantial improvement over the configurations where either layer is absent. Similarly, for the PointNet backbone with ProtoNet, the addition of both layers improves the performance to 57.86% and 86.16% in the 5-Way 1-Shot and 5-Way 5-Shot settings, respectively. These findings underline the effectiveness of attention and SoftMax layers in enhancing the discriminative power of the models, thereby significantly boosting their classification accuracy.

In the case of DGCNN with ProtoNet and Relationnet, the impact of these layers is also noticeable, though the results vary. For some configurations, the presence of both layers does not always lead to the best outcomes, indicating that the compatibility and effectiveness of these layers may depend on the specific architecture and method used.

Comparative analysis of federated learning methods

Table 5 presents a comparative analysis of various federated learning methods, including our proposed 3DFFL, on the ModelNet40 and ShapeNet datasets for 5-way 5-shot settings. The performance metrics considered are accuracy on both datasets, convergence speed, and computational efficiency. In terms of accuracy, the FedNova⁷¹ method achieves the highest performance, with an accuracy of 75.1% on ModelNet40 and 88.3% on ShapeNet. This indicates FedNova's strong capability in handling 3D point cloud data. Our proposed 3DFFL method shows competitive performance, achieving 73.8% on ModelNet40 and 87.55% on ShapeNet. This demonstrates that our approach is highly effective and is on par with the best-performing methods. Other methods, such as FedProx⁷² and FedMC⁷³, also exhibit robust performance with accuracy scores of 73.5% and 73.0% on ModelNet40, and 86.7% and 87.1% on ShapeNet, respectively. Regarding convergence speed, the proposed 3DFFL method, along with FedProx and FedMC, converges quickly, which is advantageous for real-time applications where rapid model updates are crucial. FedNova shows moderate convergence speed, balancing between accuracy and time efficiency. In contrast, FedMA⁷⁴ exhibits slower convergence, which might be a drawback in time-sensitive scenarios.

The client nodes are pre-authenticated and recognized entities. This configuration signifies a regulated environment in which all participants are familiar with and trusted by the HQ, thus obviating the necessity for

Method	ModelNet40	ShapeNet	Convergence	Resource consumption
Proposed (3DFFL)	73.8	87.55	Fast	Low
FedProx ⁷²	73.5	86.7	Fast	High
FedMA ⁷⁴	71.9	84.6	Slow	Moderate
FedMC ⁷³	73.0	87.1	Fast	High
FedNova ⁷¹	75.1	88.3	Moderate	High

Table 5. Performance comparison of different federated learning methods on ModelNet40 and ShapeNet datasets. Significant values are in bold.

dynamic client discovery and authentication procedures. Every participant, or client, is assigned an equal segment of the dataset to ensure uniformity in data distribution. In other words, the datasets are divided into three parts, guaranteeing that each client obtains a fair and well-balanced subset for training.

In terms of computational efficiency, our 3DFFL method stands out with low resource consumption, making it suitable for deployment in environments with limited computational resources. This is particularly important for applications such as autonomous vehicles, where computational efficiency is critical. On the other hand, methods like FedProx, FedMC, and FedNova, while achieving high accuracy, require higher computational resources, which could be a limitation in resource-constrained settings. It should be noted that our proposed 3DFFL method strikes a balance between accuracy, convergence speed, and computational efficiency, making it a viable solution for federated learning in 3D point cloud classification tasks. The competitive performance and efficiency of 3DFFL highlight its potential for practical applications, particularly in scenarios where data privacy and low computational overhead are essential.

Scalability and computational requirements

Our proposed framework utilizes the PointNet++ architecture, which ensures consistent computational load by processing a fixed number of points (e.g., 512 points) regardless of the initial input size. Regardless of whether the input consists of 128, 256, 512, 1024, or 2048 points, PointNet++ abstracts the points to this fixed size. This design guarantees predictable and reliable performance, with 4.07 Giga FLOPs and 1.76M parameters. In the context of federated learning, each client node performs local training on its dataset and only transmits model updates to the central server. This decentralized approach reduces the need for centralized data storage and preserves data privacy, but it does introduce communication overhead during model aggregation. The constant computational load of the PointNet++ architecture enhances resource management efficiency, making our framework particularly suitable for federated learning environments where data privacy and optimal resource utilization are crucial.

Potential limitations of the proposed approach

Despite the promising results exhibited by our proposed 3DFFL method, it is imperative to acknowledge several potential limitations. The grid search method employed for hyperparameter tuning, while exhaustive, necessitates substantial computational resources and time, resulting in a significant investment. Concerns regarding scalability may arise as the federated learning framework could face bottlenecks when applied to larger or more intricate datasets. While the federated approach guarantees data privacy, it also introduces communication overhead and heavily relies on the quality of local data, potentially impacting the overall performance of the global model.

Conclusion

Our comprehensive study introduces a pioneering approach in 3D Federated Few-Shot Learning (3DFFL), specifically tailored for the classification of 3D point clouds. The proposed method successfully integrates Federated Learning with Few-Shot Learning techniques, utilizing PointNet++ and ProtoNet as key components. This integration addresses significant challenges in the field, particularly the scarcity of labeled data and the need for privacy-preserving collaborative training. The experimental results on benchmark datasets such as ModelNet40, ShapeNet, and ScanObjectNN validate the effectiveness of our approach. Our method demonstrates competitive performance compared to state-of-the-art models, achieving high accuracy in both 5-way 1-shot and 5-way 5-shot settings. Notably, the federated aspect of the model ensures data privacy and collaborative learning among various entities, crucial in real-world applications. The success of this study not only advances the field of 3DFFL but also opens new avenues for future research. Future directions include exploring more sophisticated federated aggregation techniques to improve model robustness and accuracy, integrating advanced 3D point cloud processing methods such as graph-based neural networks or voxel-based approaches for additional performance gains, and addressing communication overhead to optimize the federated learning process for large-scale and diverse datasets. It highlights the potential of federated and few-shot learning techniques in handling the unique challenges of 3D point cloud data, setting a precedent for further exploration and development in this domain. The implications of this work extend to various applications, remarkably in the development of autonomous vehicles and other areas where 3D data processing is essential.

Data availability

The datasets utilized in this study, specifically ModelNet40, ShapeNet, and ScanObjectNN, are publicly accessible. ModelNet40 can be downloaded from the official Princeton ModelNet project website <http://modelnet.cs.princeton.edu/>, while ShapeNet can be accessed through the ShapeNet project's website <https://www.shapenet.org>. ScanObjectNN can be found at <https://hkust-vgd.github.io/scanobjectnn/>.

Code availability

The code used in this study is available at the following GitHub repository <https://github.com/raja21068/3DFFL/>.

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Author contributions

R.K. conceived the study, developed the methodology, managed project administration, and secured funding. A.A.K. executed 3D point cloud experiments, performed data analysis, and led the drafting and revisions of the manuscript. K.M.A. and S.S. contributed to the experiment design, implementation, and manuscript writing, and both reviewed and approved the final manuscript. Waqas Amin helped in the revision process and conducted additional experiments recommended during the revisions.

Competing interests

The authors declare no competing interests.

Additional information

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